LabTech Testing Limited

STANDARD TEST PROCEDURE

ASTM C939
Flow of Grout for Preplaced-Aggregate Concrete
(Flow Cone Method)

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REVISION LEDGER

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1 Scope

1.1 This test method covers a procedure for determining the time of efflux of a specified volume of fluid hydraulic cement grout through a standardized flow cone.

1.2 It is for use with neat grout and with grouts containing fine aggregate all passing a 2.36 mm sieve.

1.3 This test method is intended for use with grout having an efflux time of 35 s or less.

1.4 This test method is applicable to the determination of the fluidity of various fluid grout mixtures.

2 Equipment

2.1 Flow Cone: with dimensions as shown in Figure 1.

2.2 Receiving Container: minimum capacity 2000 mL.

2.3 Ring Stand or other device: capable of supporting the flow cone in a vertical, steady position over the receiving container.

2.4 Sprit Level.

2.5 Stop Watch: least reading of not more than 0.2 sec.

2.6 Thermometer: readable to at 0.1℃.

3 Test Sample

3.1 The grout test sample shall be in excess of 1725 mL.

3.2 The temperature of the dry materials and mixing water shall be such that the temperature of the freshly mixed grout is 23.0 °C ±2.0 °C.

4 Calibration of Apparatus

4.1 Mount the flow cone firmly in such a manner that it is free of vibration. Level the top to assure verticality.

4.2 Close the outlet of the discharge tube with a finger or a stopper. Introduce 1725 mL±5 mL of water into the cone. Adjust the point gage to indicate the level of the water surface. Then allow the water to drain.

4.3 Before first use of the flow cone with grout and periodically thereafter, check the accuracy of the cone by filling it with water as described in 4.11. After checking or adjusting the point gage, start the stop watch and simultaneously remove the finger. Stop the watch at the first break in the continuous flow of water.

4.4 The time indicated by the stop watch is the time of efflux of water. If this time is 8.0±0.2 s, the cone may be used for determining the time of efflux of grout.

5 Testing Procedure

5.1 Measure of the ambient and grout sample temperature of the start of the test.

5.2 Moisten the inside of the flow cone by filling the cone with water and, 1 minute before introducing the grout sample, allow the water to drain.

5.3 Close the outlet of the discharge tube with a finger or a stopper. Introduce the grout into the cone until the grout surface rises to contact the point gage.

5.4 Start the stop watch, and simultaneously remove the finger or stopper. Stop the watch at the first break in the continuous flow of grout from the discharge tube.

5.5 If the grout has passed sufficiently, such that light is visible through the discharge tube with borderline consistency, make sure that the fluidity of grouts meets the required specifications, the time indicated by the stop watch is the time of efflux of the grout. If light is not visible through the discharge tube, then the use of the flow cone is not applicable for grout of this consistency.

5.6 Repeat the procedures two times and make sure that at least two tests having times of efflux within 1.8 seconds of their average.

5.7 The test for time of efflux shall be made within 1 min of drawing of the grout from the mixer or transmission line. When grout is being placed over a significant period of time, the time of efflux may be determined at selected intervals to demonstrate that the consistency is suitable for the work.

6 Record

6.1 Worksheet: FST05-GTT-C939

6.2 Test Report template: GTT-C939-RT

6.3 Daily Record: FQS09-GTT-C939

7 Report

7.1 Information provided by customer.

7.2 Identification of grout sample.

7.3 Identification of materials in the grout sample.

7.4 Material proportion for field production mix.

7.5 Average time of efflux to nearest 0.2 second.

7.6 Time interval from completion of mixing.

7.7 Ambient and grout sample temperature at the beginning of the test.

8 Calibration Requirements

8.1 Flow Cone

8.1.1 Check apparatus in accordance with ASTM C939 Clause 8 using a calibrated time measuring device

8.1.2 Frequency: 1 year

8.2 Stopwatch

8.2.1 Minimum increment is 0.2 second

8.2.2 Calibrated against reference time standard

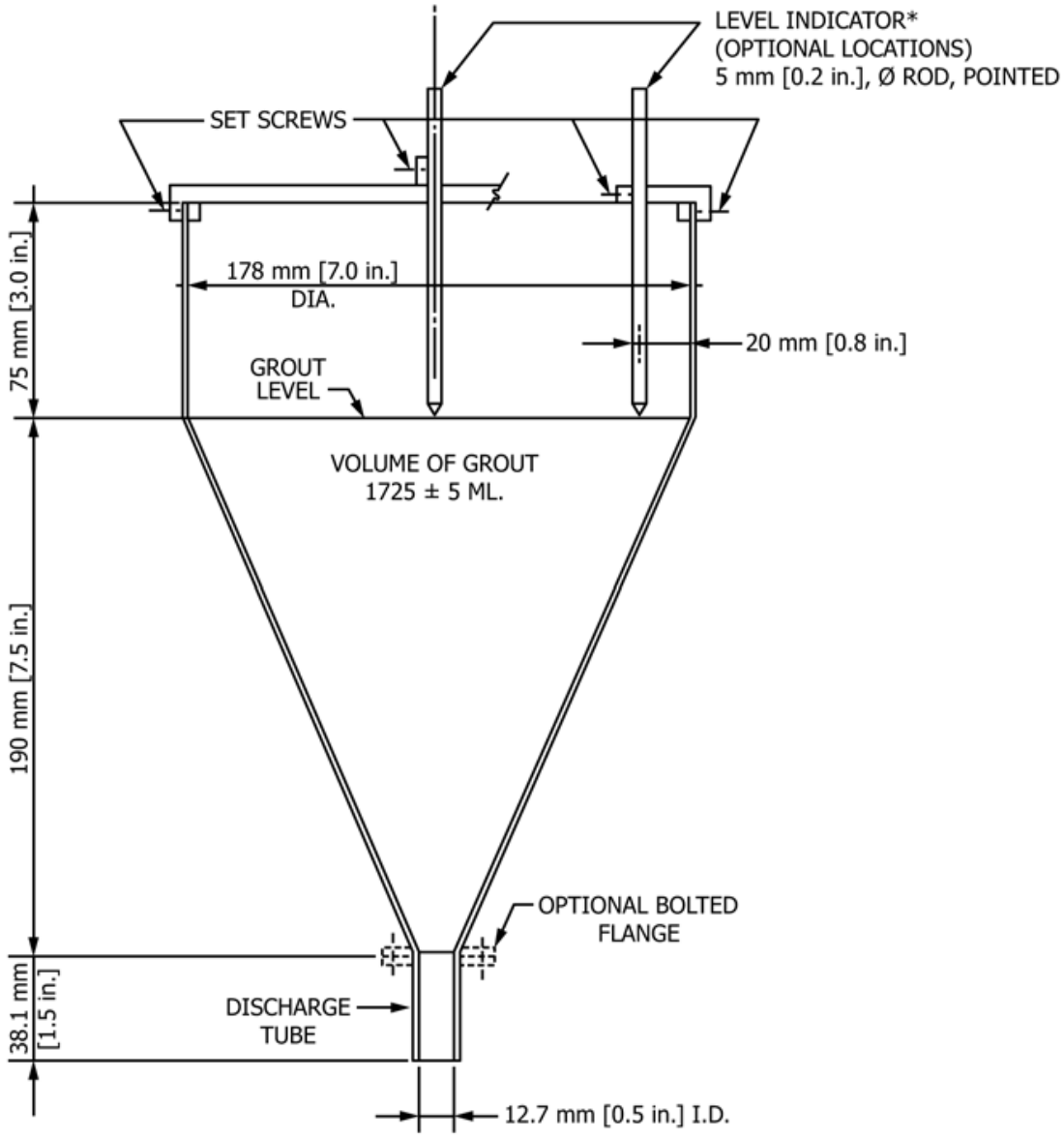
8.2.3 Frequency: 1 year

8.3 Electronic Thermometer

8.3.1 Frequency: Three years(complete), Six months(one-point)

Figure 1

Cross Section of Flow Cone



End of Procedure